

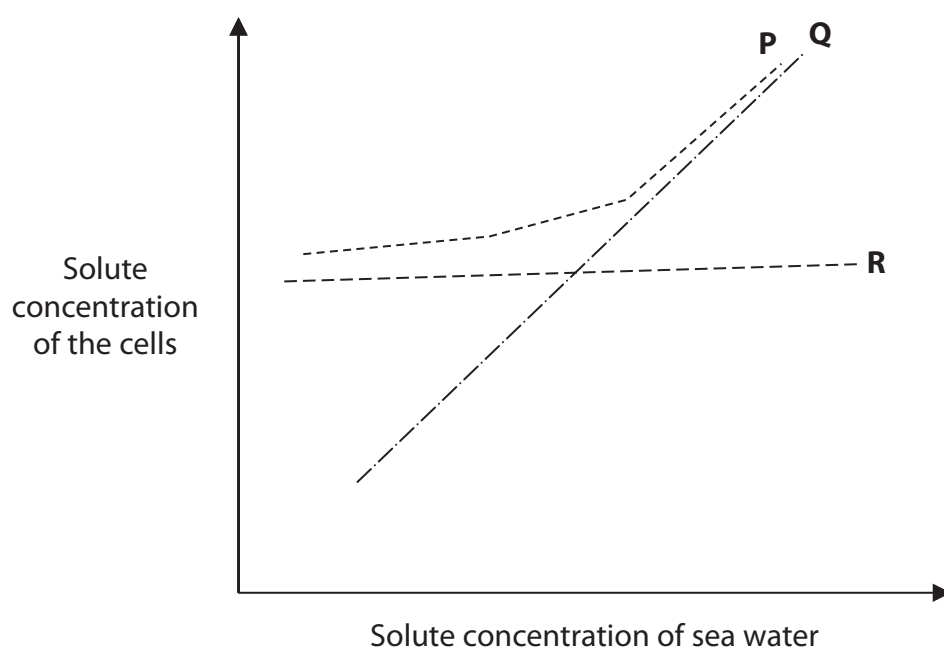
9 Some marine animals are osmoconformers and some are osmoregulators.

Strict osmoconformers have a cell solute concentration that matches the solute concentration of the sea water.

Partial osmoconformers have a cell solute concentration that varies with the solute concentration of the sea water.

Osmoregulators control the solute concentration of their cells when the solute concentration of sea water changes.

- (a) The graph shows the solute concentration of the cells of three marine animals, P, Q and R.



- (i) Which row in the table identifies the marine animals P, Q and R?

(1)

	P	Q	R
☒ A	partial osmoconformer	osmoregulator	strict osmoconformer
☒ B	partial osmoconformer	strict osmoconformer	osmoregulator
☒ C	osmoregulator	strict osmoconformer	partial osmoconformer
☒ D	strict osmoconformer	partial osmoconformer	osmoregulator

- (ii) Explain the advantages for osmoconformers of having cells with solute concentrations similar to that of the sea water they live in.

(3)

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- (b) The table shows the concentration of ions in sea water and in some osmoconformers.

Location		Ion concentration / a.u.				
		sodium	magnesium	calcium	potassium	chloride
Sea water		478	55	11	10	558
Cells of osmoconformers	jellyfish	474	53	10	11	
	sea urchin	474	54	11	10	557
	lobster	541	9	12	8	552

- (i) Estimate the chloride concentration of the cells of a jellyfish.

(1)

Answer a.u.



- (ii) Calculate the percentage difference in the concentration of magnesium ions in the lobster compared with the sea urchin.

Give your answer to two decimal places.

(2)

Answer%

- (iii) These three animals are osmoconformers even though they have different concentrations of each ion in their cells.

Explain why these three animals can be described as osmoconformers.

(2)

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- (iv) Suggest why osmoconformers do not have the same concentration of each ion in their cells.

(1)

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- (v) Energy is needed to maintain the concentration of ions inside the osmoconformers.

Explain what this energy is being used for.

(2)

(Total for Question 9 = 12 marks)

TOTAL FOR PAPER = 80 MARKS

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